

UNIT-1 Formulas

→ Taylor Series for one variable:

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^n}{n!} f^n(a)$$

about point $x=a$.

→ Maclaurin's Series for one variable:

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^n(0)$$

→ Newton Raphson Method for one variable:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

→ Algebraic Equations:

Any expression containing algebraic terms like x, x^2, x^3 is called an algebraic function.

Eg: $x^3 - x^2 + 9 = 0$.

→ Transcendental Equation:

Any expression containing exponential terms like $\log, \sin$, etc is called a transcendental function.

Eg: $xe^x - 9 = 0$, $x \sin x + \log x = 0$

→ Intermediate Value Property:

If $f(x)$ is a continuous function in $[a, b]$ and $f(a), f(b)$ are of opposite signs [$f(a) > 0, f(b) < 0$ or $f(a) < 0, f(b) > 0$] then the equation $f(x) = 0$ has atleast one real root 'c' in the interval $[a, b]$ such that $f(c) = 0$.



→ Taylor Series for function of 2 variable:

$$f(x, y) = f(a, b) + \frac{1}{1!} [(x-a) f_x(a, b) + (y-b) f_y(a, b)] + \frac{1}{2!} [(x-a)^2 f_{xx}(a, b) + 2(x-a)(y-b) f_{xy}(a, b) + (y-b)^2 f_{yy}(a, b)] + \frac{1}{3!} [(x-a)^3 f_{xxx}(a, b) + 3(x-a)^2(y-b) f_{xxy}(a, b) + 3(x-a)(y-b)^2 f_{xyy}(a, b) + (y-b)^3 f_{yyy}(a, b)] + \dots$$

→ Maclaurin's Series for function of 2 variables:

$$f(x, y) = f(0, 0) + \frac{1}{1!} [x f_x(0, 0) + y f_y(0, 0)] + \frac{1}{2!} [x^2 f_{xx}(0, 0) + 2xy f_{xy}(0, 0) + y^2 f_{yy}(0, 0)] + \frac{1}{3!} [x^3 f_{xxx}(0, 0) + 3x^2 y f_{xxy}(0, 0) + 3xy^2 f_{xyy}(0, 0) + y^3 f_{yyy}(0, 0)] + \dots$$

→ N-R Method for non-linear simultaneous equation:

$$-f(x_0, y_0) = h f_x(x_0, y_0) + K f_y(x_0, y_0)$$

$$x_1 = x_0 + h$$

$$-g(x_0, y_0) = h g_x(x_0, y_0) + K g_y(x_0, y_0)$$

$$y_1 = y_0 + K$$

→ Maxima and Minima of 2 variables: (extreme values)

$$f(x, y) = 0 \quad f_x = 0 \quad f_y = 0 \quad \rightarrow \text{to get critical points. (stationary points)}$$

$$r = f_{xx} \quad s = f_{xy} \quad t = f_{yy}$$

if,

$$rt - s^2 > 0 \quad \text{and} \quad r > 0 \quad \rightarrow \text{minimum} \quad \oplus \quad rt - s^2$$

$$rt - s^2 > 0 \quad \text{and} \quad r < 0 \quad \rightarrow \text{maximum}$$

$$rt - s^2 < 0 \quad \text{then, saddle point (neither maxima/minima)}$$

$$rt - s^2 = 0 \quad \text{then, further investigation (test fail)}$$

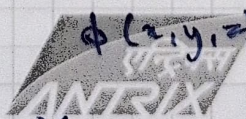
→ Lagrange's Method:

$$f = u + \lambda(\phi)$$

λ = constant

$u(x, y, z)$ = independent real variables.

$$\phi(x, y, z) = c$$



UNIT - 2 Formulas

1. Linear in 'y':

$$\frac{dy}{dx} + f(x) \cdot y = g(x)$$

Linear in 'x':

$$\frac{dx}{dy} + f(y) \cdot x = g(y)$$

2. Integrating Factor:

$$y(I.F.) = \int g(I.F.) \cdot dx + C \quad I.F. = e^{\int p \cdot dx}$$

$$I.F. = e^{\int p \cdot dy}$$

3. General Solution:

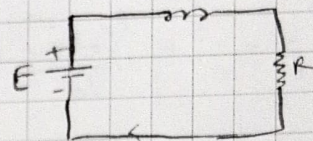
$$y(I.F.) = \int g(I.F.) \cdot dx + C$$

$$x(I.F.) = \int g(I.F.) \cdot dy + C$$

4. Application 1: Simple Electric Circuits: $L = L \frac{di}{dt}$, $C = \frac{q}{c}$, $R = Ri$

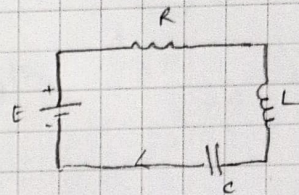
i) L-R Circuit:

$$L \cdot \frac{di}{dt} + Ri = E \Rightarrow \frac{di}{dt} + \frac{R}{L} i = \frac{E}{L}$$



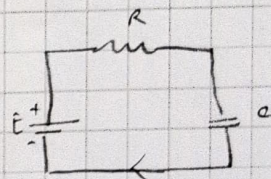
ii) L-C-R Circuit:

$$L \frac{di}{dt} + Ri + \frac{q}{C} = E \Rightarrow \frac{di}{dt} + \frac{R}{L} i + \frac{q}{LC} = \frac{E}{L}$$



iii) R-C Circuit:

$$iR + \frac{q}{C} = E \Rightarrow R \frac{dq}{dt} + \frac{q}{C} = E \Rightarrow \frac{dq}{dt} + \frac{q}{CR} = \frac{E}{R}$$



5. Application 2: Newton's Law of Cooling.

change in temperature of body is directly proportional to difference in the temperature of body and surrounding medium.

$$\frac{dT}{dt} \propto (T - t_2)$$

T = temp of body at any time.

t_1 = initial temp of body.

t_2 = surrounding temp.

Sol.

$$T = t_2 + (t_1 - t_2) e^{-kt}$$



Application 3: Orthogonal Trajectories

if 2 families of curves are such that every member of 1 family intersects with every member of the other family at right angles then they are OT.

Procedure: i) given $f(x, y, c) = 0$, diff w.r.t x and eliminate c .

(for cartesian) ii) replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$

iii) separate the variables and simplify.

Procedure: i) given $f(r, \theta, c) = 0$, diff w.r.t θ and eliminate c . Find $\frac{dr}{d\theta}$

(for polar) ii) replace $\frac{dr}{d\theta}$ with $-r^2 \frac{d\theta}{dr}$

iii) separate the variable and simplify.

7. Taylor Series Method: $y' = \frac{dy}{dx} = f(x, y)$

$$y = y(x_0) + y'(x_0) \frac{(x-x_0)}{1!} + y''(x_0) \frac{(x-x_0)^2}{2!} + y'''(x_0) \frac{(x-x_0)^3}{3!} + \dots$$

8. Euler's Method:

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$h = \text{step length.}$

$$\frac{dy}{dx} = f(x, y)$$
$$y(x_0) = y_0$$

9. Modified Euler's Method:

$$y_1^{(0)} = y_0 + hf(x_0, y_0)$$

$$y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})]$$

10. Runge - Kutta Method:

$$\frac{dy}{dx} = f(x, y) \quad y(x_0) = y_0$$

calculate: $k_1 = hf(x_0, y_0)$

$$k_2 = h[f(x_0 + h/2, y_0 + k_1/2)]$$

$$k_3 = h[f(x_0 + h/2, y_0 + k_2/2)]$$

$$k_4 = h[f(x_0 + h, y_0 + k_3)]$$

$$\therefore y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$



UNIT - 3

Formulas

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

→ Homogeneous - if $\phi(x) = 0$
→ Non-Homogeneous - $\phi(x) \neq 0$

1. Homogeneous LDE:

$$CF = y$$

i) if m_1, m_2 real and distinct

$$y_c = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

ii) if roots are real and equal

$$y_c = (C_1 + C_2 x + C_3 x^2 + \dots) e^{mx}$$

iii) if $m = p \pm iq$

$$y_c = e^{px} (C_1 \cos qx + C_2 \sin qx)$$

2. Non-Homogeneous LDE:

$$y = y_c + y_p$$

$$PI = \frac{1}{f(D)} \cdot \phi(x)$$

→ Type 1:

$$y_{PI} = \frac{1}{f(D)} \cdot e^{ax}$$

put $(D=a)$

$$\rightarrow 2 \cos x \cdot \sin y = \sin(x+y) - \sin(x-y)$$

$$\rightarrow 2 \sin x \cdot \cos y = \sin(x+y) + \sin(x-y)$$

$$\rightarrow 2 \cos x \cdot \cos y = \cos(x+y) + \cos(x-y)$$

$$\rightarrow 2 \sin x \cdot \sin y = \cos(x+y) - \cos(x-y)$$

→ Type 2:

$$y_p = \frac{1}{f(D)} \cdot \sin(ax) / \cos(ax)$$

put $(D^2 = -a^2)$

$$\rightarrow \sin^2 x = \frac{1 - \cos^2 x}{2}$$

$$\rightarrow \cos^2 x = \frac{1 + \cos^2 x}{2}$$

→ Type 3:

$$y_p = \frac{1}{f(D)} \cdot x^m$$

(polynomial)

$$\rightarrow (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$\rightarrow (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

$$\rightarrow (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$\rightarrow (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$



→ Type 4:

$$y_p = \frac{1}{f(D)} \cdot e^{ax} \cdot v$$

$$= e^{ax} \cdot \frac{1}{f(D)} \cdot v$$

→ Type 5:

$$y_p = \frac{1}{f(D)} \cdot x^m \cdot v$$

$$= \left(x^m - \frac{f'(D)}{f(D)} \right) \frac{v}{f(D)}$$

$$\rightarrow \int u v dx = uv_1 - u'v_2 + u''v_3 \dots$$

$$\rightarrow \int \tan x dx = \log (\sec x) + c$$

$$\rightarrow \int \sec x dx = \log (\sec x + \tan x) + c$$

$$\rightarrow \int \cot x dx = \log (\sin x) + c$$

$$\rightarrow \int \operatorname{cosec} x dx = \log (\operatorname{cosec} x - \cot x)$$

3. Initial Value Problem: (IVP)

one in which a sol. to an equation is obtained to conditions on the unknown function and its derivatives specified at one value of independent variable. This is called initial condition.

eg: $y'' + 2y' + y - 6y = 0$ $y(0) = -1$ $y'(0) = 1$ $y''(0) = 2$.

4. Boundary Value Problems: (BVP)

one in which a sol. to an equation is obtained to conditions on the unknown function and its derivatives specified at 2 or more values of independent variable is a boundary cond.

eg: $(D^2 + 2D - 2)y = e^{2x}$ $y(0) = 1$ $y(1) = 2$.

5. Method of Variation of Parameters: (MVP)

$$a_0 y''' + a_1 y'' + a_2 y' + a_3 y = \phi(x)$$

$$y_c = C_1 y_1 + C_2 y_2$$

$$\omega = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$y_p = A y_1 + B y_2$$

$$A = - \int \frac{y_2 \phi(x)}{\omega} dx$$

$$B = \int \frac{y_1 \phi(x)}{\omega} dx$$



6. Cauchy's LDE:

$$a_0 x^3 y''' + a_1 x^2 y'' + a_2 x y' + a_3 y = \phi(x)$$

$$\text{put } t = \log x$$

$$x = e^t$$

③ if $a=1$ and $b=0$ in Legendre's L.D.E then it reduces to Cauchy's linear diff eq.

7. Legendre's LDE:

$$a_0 (ax+b)^3 y''' + a_1 (ax+b)^2 y'' + a_2 (ax+b) y' + a_3 y = \phi(x)$$

$$t = \log(ax+b)$$

$$ax+b = \log t$$

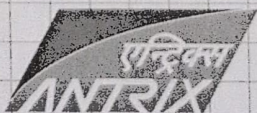
④ for Cauchy's

$$x y' = D y$$

$$x^2 y'' = D(D-1) y$$

$$x^3 (y''') = D(D-1)(D-2) y$$

_____ X _____



UNIT - 4 Formulas

1. Newton's Forward Interpolation Formula:

$$y(x) = y_0 + p \cdot \Delta y_0 + p \frac{(p-1)}{2!} \cdot \Delta^2 y_0 + p \frac{(p-1)(p-2)}{3!} \cdot \Delta^3 y_0 + \dots \\ + p \frac{(p-1)(p-2) \dots (p-(n-1))}{n!} \cdot \Delta^n y_0$$

where $p = \frac{x - x_0}{h}$

$\Rightarrow h =$ equally spaced values.

2. Newton's Backward Interpolation Formula:

$$y(x) = y_n + p \cdot \nabla y_n + p \frac{(p+1)}{2!} \cdot \nabla^2 y_n + p \frac{(p+1)(p+2)}{3!} \cdot \nabla^3 y_n + \dots \\ + p \frac{(p+1)(p+2) \dots (p+(n-1))}{n!} \cdot \nabla^n y_n$$

where $p = \frac{x - x_n}{h}$

$h =$ eq. spaced values.

3. Lagrange's interpolation formula:

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3) \dots (x_0-x_n)} \cdot y_0 + \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} \cdot y_1 \\ + \dots + \frac{(x-x_0)(x-x_1)(x-x_3) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} \cdot y_n$$

4. Inverse - Lagrange's Interpolation Formula:

$$x(y) = \frac{(y-y_1)(y-y_2) \dots (y-y_n)}{(y_0-y_1)(y_0-y_2)(y_0-y_3) \dots (y_0-y_n)} \cdot x_0 + \frac{(y-y_0)(y-y_2) \dots (y-y_n)}{(y_1-y_0)(y_1-y_2) \dots (y_1-y_n)} \cdot x_1 \\ + \dots + \frac{(y-y_0)(y-y_1) \dots (y-y_{n-1})}{(y_n-y_0)(y_n-y_1) \dots (y_n-y_{n-1})} \cdot x_n$$

5. Newton's divided difference formulas

$$y(x) = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] + (x-x_0)(x-x_1)(x-x_2)[x_0, x_1, x_2, x_3] + \dots + (x-x_0)(x-x_1) \dots (x-x_{n-1})[x_0, x_1, \dots, x_n]$$

where $[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_1}$$



Numerical differentiation using Newton's forward interpolation:

$$i) \left(\frac{dy}{dx} \right)_{x=x_0+ph} = \frac{1}{h} \left[\Delta y_0 + \frac{(2p-1)}{2} \Delta^2 y_0 + \frac{(3p^2-6p+2)}{6} \Delta^3 y_0 + \frac{(4p^3-18p^2+22p-6)}{24} \Delta^4 y_0 + \dots \right]$$

$$ii) \left(\frac{d^2y}{dx^2} \right)_{x=x_0+ph} = \frac{1}{h^2} \left[\Delta^2 y_0 + (p-1) \Delta^3 y_0 + \frac{(12p^2-36p+22)}{24} \Delta^4 y_0 + \dots \right]$$

$$\text{where } p = \frac{(x-x_0)}{h}$$

~~$$*) \left(\frac{dy}{dx} \right)_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{2}{6} \Delta^3 y_0 - \frac{6}{24} \Delta^4 y_0 + \dots \right]$$~~

using $p=0$ in the above 2 we get:

$$ii) \left(\frac{dy}{dx} \right)_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{2}{6} \Delta^3 y_0 - \frac{6}{24} \Delta^4 y_0 + \dots \right]$$

$$iv) \left(\frac{d^2y}{dx^2} \right)_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{22}{24} \Delta^4 y_0 + \dots \right]$$

7. ~~Newton's~~ Numerical differentiation using Newton's backward interpolation:

$$i) \left(\frac{dy}{dx} \right)_{x=x_n+ph} = \frac{1}{h} \left[\nabla y_n + \frac{2(p+1)}{2!} \nabla^2 y_n + \frac{(3p^2+6p+2)}{6} \nabla^3 y_n + \frac{(4p^3+18p^2+22p+6)}{24} \nabla^4 y_n + \dots \right]$$

$$ii) \left(\frac{d^2y}{dx^2} \right)_{x=x_n+ph} = \frac{1}{h^2} \left[\nabla^2 y_n + (p+1) \nabla^3 y_n + \frac{(12p^2+36p+22)}{24} \nabla^4 y_n + \dots \right]$$

$$\text{where } p = \frac{(x-x_n)}{h}$$

using $p=0$ in the above 2 eq. we get.

$$iii) \left(\frac{dy}{dx} \right)_{x=x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{2}{6} \nabla^3 y_n + \frac{6}{24} \nabla^4 y_n + \dots \right]$$

$$iv) \left(\frac{d^2y}{dx^2} \right)_{x=x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{22}{24} \nabla^4 y_n + \dots \right]$$



Radius of curvature formula:

$$(r) \text{ at } x = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

9. Numerical Integration:

here $h = \frac{b-a}{n}$ where, $a = \text{lower limit}$ $b = \text{upper limit}$

i) Trapezoidal Rule:

$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

used when interval is $n = 1, 2, 3, \dots$

ii) Simpson's 1/3rd rule:

$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + y_5 \dots) + 2(y_2 + y_4 + y_6 \dots)]$$

used when interval is $n = 2, 4, 6, \dots$

iii) Simpson's 3/8th rule:

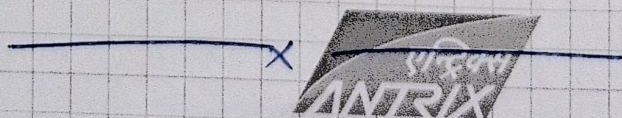
$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + y_6) + 2(y_3 + y_7 + y_8 \dots)]$$

used when interval is $n = 3, 6, 9, \dots$

⊛ most approximate answer: Simpson's 1/3rd rule.

⊛ if ordinates are mentioned then sub-interval n is less than one ordinate, eg. ordinate = 8.

intervals $n = 7$.



UNIT - 5

Linear Algebra.

→ Elementary Row Operations:

$a_{11}, a_{22}, a_{33} \rightarrow$ each leading entry is ⁺ to the right of leading entry of previous row.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

* All zero rows are below non-zero rows.

* first non zero element is called leading entry.

→ Rank of a Matrix: (ρ)

$$\rho(A) = \rho(A^T)$$

rank of matrix in Echelon form is equal to no. of non-zero rows.

→ Consistency of Solution: ($AX = B$)

1. Unique Solution = $\rho(A) = \rho(A:B) = \text{number of variables}$.
2. infinite solution = $\rho(A) \neq \rho(A:B) \neq \text{number of variables}$.

→ Homogeneous System: ($AX = 0$)

1. Trivial Solution: $XI = 0$
2. Non-Trivial Solution: coefficient = 0

→ Gauss - elimination Method:

equation $\rightarrow$ matrix $\rightarrow$ echelon form $\rightarrow$ ~~find~~ equation $\rightarrow$ find x, y, z .

→ Gauss - Siedel method:

(first check if diagonally dominant)

$$x_1' = \frac{(b_1 - a_{12}x_2^0 - a_{13}x_3^0)}{a_{11}}$$

$$x_2' = \frac{(b_2 - a_{21}x_1' - a_{23}x_3^0)}{a_{22}}$$

$$x_3' = \frac{(b_3 - a_{31}x_1' - a_{32}x_2')}{a_{33}}$$

assume initial as x_1^0, x_2^0, x_3^0

First Iteration.



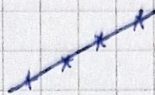
⊗ if values are not given $(0, 0, 0)$.

Geometrical Meaning:

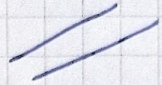
1. unique solution



2. Infinite solution



3. No solution



→ Eigen Values and Eigen Vectors:

in a square matrix if there exists a scalar ^{constant} ~~constant~~ λ and a non-zero matrix such that $AX = \lambda X$.

$\lambda \rightarrow$ is eigen value of matrix A.

$X \rightarrow$ is eigen vector of matrix A.

$$(A - \lambda I)X = 0.$$

$$\begin{bmatrix} a_1 - \lambda & b_1 & c_1 \\ a_2 & b_2 - \lambda & c_2 \\ a_3 & b_3 & c_3 - \lambda \end{bmatrix}$$

→ Rayleg's Power Method:

determines largest eigen vector of square matrix.

$$AX^{(0)} = \lambda^{(1)} X^{(1)}$$

$$AX^{(1)} = \lambda^{(2)} X^{(2)}$$

→ Diagonalization of Matrices:

- i) Modal matrix: matrix which diagonalizes given square matrix. (P)
- ii) Spectral matrix: in process of diagonalization resultant matrix. (D)
- iii) Similarity Transformation: in diagonal matrix D, diagonal elements are eigen values of A. This process is st.
 $(P^{-1}AP = D)$

→ Application of first order / degree ODE's using Matrix Method:

$$X = P \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a c_1 e^t + c_2 e^t \\ b c_1 e^t + d c_2 e^t \end{bmatrix}$$

replace values of

x_1, x_2 given to find c_1, c_2 . then find t.



PROPERTIES OF EIGEN VALUES

① Any Square matrix A and A^T/A' have the same eigen values.

Proof: let A be a square matrix.

$$(A \pm B)^T = A^T \pm B^T$$

$$(A - \lambda I)^T = A^T - \lambda I^T$$

$$(A - \lambda I)^T = A^T - \lambda I$$

$$\textcircled{1} |(A - \lambda I)^T| = |A^T - \lambda I| \quad (|A^T| = |A|)$$

$$\textcircled{2} |A - \lambda I| = |(A - \lambda I)^T|$$

characteristic eq. $|A - \lambda I| = 0$

$$\therefore |A - \lambda I| = |A^T - \lambda I| = 0$$

$\therefore A$ and A^T have same eigen values.

③ The eigen value of a triangular matrix are the diagonal elements.

Proof: let $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & & \\ \vdots & \vdots & a_{33} & \\ & & & a_{nn} \end{bmatrix}$

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ 0 & a_{22} - \lambda & & \\ \vdots & \vdots & a_{33} - \lambda & \\ & & & a_{nn} - \lambda \end{vmatrix} = 0$$

$$(a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) \dots (a_{nn} - \lambda) = 0$$

$$\left. \begin{array}{l} a_{11} - \lambda = 0 \\ a_{22} - \lambda = 0 \end{array} \right\} a_{nn} - \lambda = 0$$

$$* \lambda = a_{11}, a_{22}, a_{33}, \dots, a_{nn} *$$

$\therefore a_{11}, a_{22}, a_{33}, \dots$ are the roots of $|A - \lambda I| = 0$.

② If λ is an eigen value of matrix A then $1/\lambda$ is eigen value of A^{-1} .

Proof: let A be square matrix.

if λ is eigen of matrix A then,

$$AX = \lambda X \quad \dots \textcircled{1}$$

$$A^{-1}(AX) = A^{-1}(\lambda X)$$

$$(A^{-1}A)X = \lambda(A^{-1}X) \quad (A^{-1}A) = I$$

$$IX = \lambda(A^{-1}X)$$

$$\therefore A^{-1}X = \frac{1}{\lambda}X \quad \dots \textcircled{2}$$

from 1 and 2,

$\frac{1}{\lambda}$ is the eigen value of matrix A^{-1} .

④ $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigen values of mA , then $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ are eigen values of A^m , $m \in \mathbb{I}^+$

Proof: let A be square m , λ be eigen value

then, $AX_i = \lambda X_i$ multiply both by A .

$$A(AX_i) = (\lambda X_i)A$$

$$A^2 X_i = \lambda (AX_i)$$

$$A^2 X_i = \lambda (\lambda X_i) \quad \lambda^2 (X_i)$$

$$A^2 X_i = \lambda^2 X_i$$

$$A^3 X_i = \lambda^3 X_i$$

$$\therefore A^m X_i = \lambda^m X_i$$

$\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ are eigen values of A^m .